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1. The Community Problem Analysis

1.1 The Problem description: Flooding

Flooding is a well-known hazard that has a major annual impact on the people and infrastructure of Sierra Leone. Year after year, its occurrence has consistently put residents, homes, roads, schools, health facilities, and other critical services at serious risk.

When civilians do not receive clear and timely information about rising water levels or potentially affected areas, they may have difficulties or delays in taking precautions or deciding where to seek help.

1.2 Who is affected?

- Civilians residing in flood-prone communities, especially those who may need of urgent evacuation.
- Local emergency responders and community leaders who may be affected by the flood themselves and will need advance reports on affected areas to make precautionary decisions (such as evacuation)
- Schools, clinics, businesses, and transport services that may be disrupted directly or indirectly by flooding.

1.3 Why the Problem Exists (Background)

Many of the flooding disasters that have destroyed lives and property in Sierra Leone can be linked to poor land-use planning, a lack of building permits, and extensive deforestation, according to Mayor Aki-Sawyer (Davies, 2023, para. 4). Sierra Leone is no stranger to torrential rainfalls: between July and September each year, seasonal rains cause localized flooding that affects countless families, often with serious health and socioeconomic consequences (World Health Organization Regional Office for Africa, 2016, para. 2).

These problems are unfortunately the result of a mix of civilian carelessness and government negligence. Even before the tragic August 2017 incident, which left 3,000 people homeless and over 1,000 dead (The Conversation, 2022), civilians had been warned that "irresponsible actions taken on the hills will affect the city greatly" (Drury, 2017, para. 18). Mohamed Bah, deputy director of Sierra Leone's Environment Protection Agency, also told the Standard Times Newspaper that such flooding would likely continue unless civilians changed their waste disposal methods and stopped clearing trees recklessly (Standard Times, 2015). Others, such as Mr. Hitchen, have pointed to a lack of political will to tackle the issue head-on, as reported by Flora Drury (Drury, 2017, para. 20).

1.4 How the Proposed Solution Addresses the Problem

One potential solution that could help better combat this hazard would be implementing a "Community Flood Alert and Response Information System" (CFARIS). Such a system would be created with the capability to collect flood reports, classify the potential risk of a flood using criteria set by local authorities, and share verified warnings and safety information to assist civilians, local emergency responders and community leaders in taking precautionary measures in case of a potential flood.

At its core, this is a simple system that accepts flood reports from residents or designated community leaders. It checks reports for any necessary information pertaining to potential floods (such as rising water levels, clogged drainages or torrential rain) in the surrounding area, flags the urgent observations, and then classifies the situation using risk criteria provided by the relevant local authorities. The CFARIS system would then share verified alerts and, where available and necessary, official safety instructions to civilians.

The system could share such updates through channels available to the community, such as SMS, radio announcements, or community notice points. Its exact channels of communication would, however, depend on local technological access and resources but by providing clear and timely information, CFARIS would help residents prepare earlier and give responders the reports they need to act ahead of time.

1.5 The community impact

Flooding is often a threat to civilians, homes, schools, health facilities, businesses, roads, and other community services. When residents lack timely information about changing conditions, they may not know when to avoid a route, prepare for potential flooding, or seek official guidance. This project therefore suggests the Community Flood Alert and Response Information System (CFARIS) as a potential solution; its purpose is to collect and organize simple observations about possible flood conditions and present a clear system assessment to the user that reflects the observations made. The questions are designed around signs a person may be to observe, such as prolonged heavy rain, overflowing drains, rising waterways, or water accumulating in low-lying areas. The system does not require the user to possess technical knowledge nor measure water depth or volume, instead CFARIS guides the user through a menu of options to submit a report, check the latest updates or back out; when choosing to submit a report, the user is prompted to provide the area and answer the system's Yes/No questions. The system checks that the required answers are present, applies its risk rules in priority order, selects the matching safety guidance, and displays the resulting assessment with its observed date/time.

The intended beneficiaries are residents who are in need of easily understandable flood-related information and community members/leaders who may use reports to stay alert to changing conditions; a clear update helps a user recognize reported warning signs and seek official guidance as necessary. It should be noted that the system is intended to support civilian awareness and preparedness; it is not capable of preventing flooding nor replace official emergency services.

2. Sustainable Development Goal (SDG) alignment

CFARIS relates to '**SDG 11: Sustainable Cities and Communities**', especially Target 11.5 which calls for reducing deaths, the number of people affected, and economic losses caused by disasters, including water-related disasters. A system that organizes and communicates local flood reports would support public awareness and awareness (United Nations Department of Economic and Social Affairs, n.d).

The system also relates to 'SDG 13: Climate Action'. Target 13.1 focuses on strengthening resilience and adapting to climate-related hazards and natural disasters; target 13.3 of the United Nations SDGs also includes improving awareness on climate impacts and providing early warnings (United Nations Department of Economic and Social Affairs, n.d). CFARIS would contribute to these Sustainable Development Goals by presenting easily understood risk information and safety guidance, provided the information is accurate, accessible, and appropriately labelled.

3. Why it should be Open-Source

Making CFARIS open source would allow students, developers, community groups, and relevant organizations to inspect how its logic works and suggest or implement improvements themselves. Others could also adapt the questions asked, the messages, and the list of locations to their local needs or simply translate the interface into languages used by their communities so as to help overcome any existing language barriers; it being public access would also make it easier to identify unclear rules or errors in its design.

4. Digital Public Good Justification and long-term sustainability

CFARIS could have a lot of potential as a digital public good if it were developed for broad public benefit, made openly available under a suitable license, documented, easily accessible, and designed with user privacy and safety protections in mind. Due to the nature of the CFARIS system, it is only natural that it promotes accessibility and openness (as it is intended to be in plain-language and made available to the public as they are the target audience) and its benefit to the public is also apparent in that it is built entirely to assist the public in taking precautionary actions prior to any potential flooding as well as alert the public of any flood-prone area and provide the necessary safety instructions.

As for long-term sustainability, in theory it should be quite sustainable long-term. While it may indirectly consume water for cooling due to its data centers potentially generating a lot of heat due to the intended scope of the system, it should by no means be as consumptive as AI data centers and should be fairly sustainable in that regard in exchange for supporting social wellbeing (alerts the users of flooding and keeps them updated on reports) and its economic viability should be quite good as it should also be fairly manageable in that regard without compromising either current or future generations, thereby allowing the CFARIS system to potentially achieve the three core pillars of long-term sustainability as explored by the Food and Agriculture Organization of the United Nations (FAO) and the Enel Learning Hub.

5. Computational thinking process and algorithm design

The system can be divided into smaller tasks that are at play: display the menu, collect a report, validate the answers, classify the reported conditions, display/retrieve an update and output the appropriate safety guidance; breaking a larger task into smaller parts such as this is an appropriate example of problem decomposition. CFARIS uses variables such as UserChoice, Location, the Yes/No observation questions, and SafetyGuidance; its decision logic applies IF/ELSE in an order of priority.

For instance, the classification logic used by the system checks 'Urgent' conditions first, followed by 'Warning' and 'Advisory' conditions; 'Safe' is assigned only if all required questions are answered and no listed risk signs are reported whereas missing or unclear answers lead to 'Unclear' rather than 'Safe'. This order of priority is essential as a serious answer of greater priority should not be overridden by a less serious one even if both answers were observed.

5.1 Functional modules

The CFARIS system's design could be divided into functional modules as well: [Validate_Report] checks required fields; [Classify_Risk] applies the risk rules and returns the appropriate risk level; [Select_Safety_Guidance] chooses the message for the assigned level; and [Display_Update] formats the information for the user. These modules make the overall system design both easier to understand for users and easier to maintain for developers.

6. Expected Outputs

6.1 Scenario 1: No immediate signs reported

Example answers:
ContinousHeavyRain==”No”
DrainOverflow==”No”
BuildingInfiltration==”No”
WaterAccumulation==”No”
ImmediateDanger==”No”
RapidRise==”No”
WaterDisruption==”No”

Expected Output:
Area: [Location]
Risk Assessment: Safe
Observation Date & Time: [ObservationDateTime]
Safety Guidance: “No immediate signs of flooding were reported but this does not guarantee safety, continue to monitor the environment and follow any official instructions"

6.2 Scenario 2: Early warning signs

Example answers:
ContinousHeavyRain==”Yes”
RapidRise==”No”
DrainOverflow==”No”
WaterAccumulation==”Yes”
BuildingInfiltration==”No”
ImmediateDanger==”No”
WaterDisruption==”No”

Expected Output:
Area: [Location]
Risk Assessment: Advisory
Observation Date & Time: [ObservationDateTime]
Safety Guidance: “Prepare to move to a safer place if conditions worsen and follow local emergency instructions if given, avoid reported flood-prone areas and affected routes"

6.3 Scenario 3: Worsening Conditions

Example answers:
ContinousHeavyRain==”No”
RapidRise==”Yes”
DrainOverflow==”Yes”
WaterAccumulation==”No”
BuildingInfiltration==”No”
ImmediateDanger==”No”
WaterDisruption==”Yes”

Expected Output:
Area: [Location]
Risk Assessment: Warning
Observation Date & Time: [ObservationDateTime]
Safety Guidance: “Avoid drains and floodwater, follow local emergency instructions and stay alert for changing conditions"

6.4 Scenario 4: Serious danger indicator

Example answers:
ContinousHeavyRain==”No”
RapidRise==”No”
DrainOverflow==”No”
WaterAccumulation==”No”
BuildingInfiltration==”Yes”
ImmediateDanger==”Yes”
WaterDisruption==”No”

Expected Output:
Area: [Location]
Risk Assessment: Urgent
Observation Date & Time: [ObservationDateTime]
Safety Guidance: “Move away from affected areas and to a safer place if you can, do not walk or drive through floodwater. If anyone is trapped, injured, or in immediate danger, contact local emergency services or authorities (Call NDMA toll-free: 1199)."

6.5 Scenario 5: Unclear Instructions

Example answers:
ContinousHeavyRain==”Maybe”
RapidRise==”I guess”
DrainOverflow==”Yeah”
WaterAccumulation==”Nope”
BuildingInfiltration==”Not Really”
ImmediateDanger==”Probably”
WaterDisruption==”Unsure”

Expected Output:
Area: [Location]
Risk Assessment: Urgent
Observation Date & Time: [ObservationDateTime]
Safety Guidance: “Information was unclear and there isn't enough information to assess the situation."

6.6 Scenario 6: Check latest update

Example answer:
UserChoice==”2”

Expected Output:
Area: [Location]
Risk Assessment: Safe
Observation Date & Time: "25 October 2026, 10:00 AM"
Safety Guidance (At time of update): No immediate signs of flooding were reported but this does not guarantee safety. Continue to monitor the environment and follow any official instructions"

6.7 Scenario 7: Exit system

Example answer:
UserChoice==”3”

Expected Output:
“You have chosen to exit.”

6.8 Scenario 8: Invalid Input

Example answer:
UserChoice==”4” or UserChoice==String

Expected Output:
"Your input was invalid, please try again next time."

7. Github Evidence:
   7.1 Github Repository:
   PROG101_Assignment1_DanielJoelDangiwa_905006536

   7.2 Repository URL:
   https://github.com/Erocrhyder/PROG101_Assignment1_DanielJoelDangiwa_905006536

   8.3 Git Commit History:

   7.3 README.md

   7.4 Repository Screenshot:

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